AP CALCULUS AB

Unit 1 • Worksheet 1-5A • Answer key

Worksheet 1-5A • IVT and EVT

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

f is continuous on $[-1, 3]$, with $f(-1) = 5$ and $f(3) = -2$. Which must be true?

(A) $f(1) = 1.5$

(B) $f(c) = 0$ for some c in $(-1, 3)$

(C) The minimum of f is -2.

(D) f is decreasing.

Solution

Answer: (B). IVT: 0 is between 5 and -2.

Question 2

NO CALCULATOR

f is continuous on $[2, 5]$, with $f(2) = 3$ and $f(5) = 11$. The IVT guarantees that $f(c)$ equals which value for some c in $(2, 5)$?

(A) 12

(B) 2

(C) 7

(D) 0

Solution

Answer: (C). Only 7 is between 3 and 11.

Question 3

NO CALCULATOR

h is continuous.

x	0	1	3	4	6
$h(x)$	-2	3	1	-1	2

What is the least number of zeros of h on $[0, 6]$?

(A) 1

(B) 2

(C) 3

(D) 4

Solution

Answer: (C). Sign changes on $[0, 1]$, $[3, 4]$ and $[4, 6]$.

Question 4

NO CALCULATOR

Which function is guaranteed to have an absolute maximum on the given interval?

(A) x^3 on $[0, 2]$
(B) $\frac{1}{x}$ on $(0, 1]$
(C) x^2 on $(-1, 1)$
(D) $\frac{1}{x-3}$ on $[2, 4]$

Solution
Answer: (A). EVT: continuous on a closed interval.

Question 5

NO CALCULATOR

Which interval matches $1 \leq x < 4$?

(A) $(1, 4)$
(B) $[1, 4]$
(C) $(1, 4]$
(D) $[1, 4)$

Solution
Answer: (D). 1 included (bracket), 4 excluded (parenthesis).

Question 6

NO CALCULATOR

What does $f(c) \leq f(x)$ for every x in $[a, b]$ mean?

(A) f has an absolute minimum at $x = c$.
(B) f has an absolute maximum at $x = c$.
(C) $f(c) = 0$.
(D) f is decreasing.

Solution
Answer: (A). The height at c is below all the others.

Question 7

NO CALCULATOR

$f(x) = \frac{1}{x}$ has $f(-1) = -1$ and $f(1) = 1$, but $f(x)$ is never 0. Why doesn't this contradict the IVT?

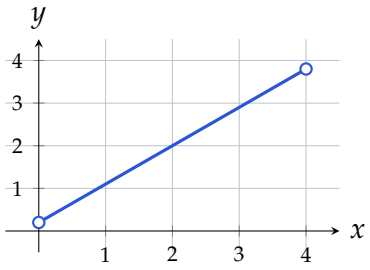
(A) 0 is not between -1 and 1.
(B) The interval is open.
(C) f is not continuous on $[-1, 1]$.
(D) The IVT only works for polynomials.

Solution
Answer: (C). f has a vertical asymptote at $x = 0$.

Question 8

NO CALCULATOR

The graph of f on the open interval $(0,4)$ is shown. Which is true?



(A) f has a maximum at $x = 4$.
(B) f has a maximum and a minimum on $(0,4)$.
(C) f has no maximum and no minimum on $(0,4)$.
(D) The EVT guarantees a maximum.

Solution

Answer: (C). The endpoints are not included, so the highest and lowest heights are never reached. The EVT needs a closed interval.

Question 9 (Free response)

NO CALCULATOR

The temperature $T(t)$, in °C, is continuous.

t (h)	0	2	5	8
$T(t)$	12	18	15	9

(a) Must $T(t) = 16$ for some t in $(0,2)$? Justify.
(b) What is the least number of times $T(t) = 16$ on $[0,8]$? Explain.
(c) Must T have an absolute maximum on $[0,8]$? What can you say about its value?

Solution

(a) Yes: T is continuous on $[0,2]$ and $12 < 16 < 18$ (IVT).
(b) At least 2: once in $(0,2)$ and once in $(2,5)$, since $18 > 16 > 15$.
(c) Yes, EVT (continuous on the closed interval $[0,8]$). The maximum is at least 18.